POLITECNICO
MILANO 1863

MPI

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Rules

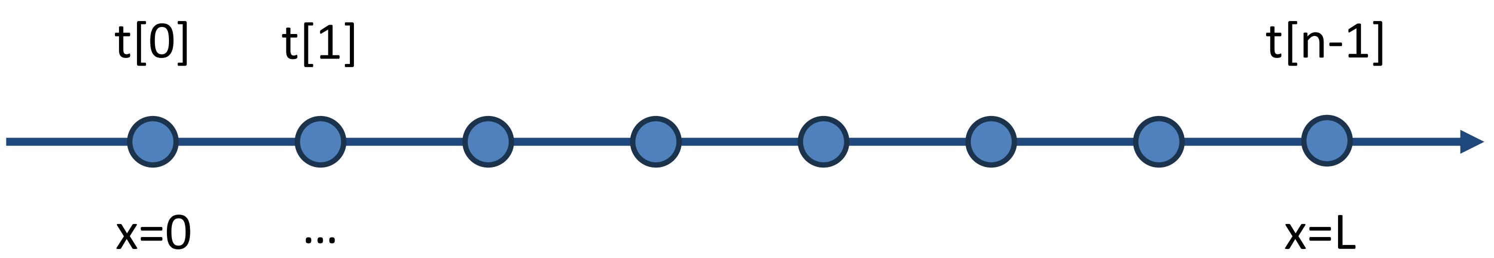
- Rename the temp_simulation_XX.c file replacing XX with the number of your group

- Write in the comment on top of the file your group number and the name of all group members

- Submit only a single c file with your solution
  - Submitted from the contact email provided in the group registration document

Requirements

- You are to implement a simple simulator of temperature dynamics using MPI
  - Consider a one-dimensional domain of length L
  - Consider the value of temperature at n discrete points within the domain



Requirements

- The simulation evolves in descrete rounds, each containing multiple iterations
  - iterations_per_round in the template file

- Before the first round, the initial temperature value of each point needs to be initialized
  - You can use the initial_condition function
  - It takes in input the position of the point (x) and the length of the domain (L)

Requirements

- Let us call $t[p]^i$ the value of temperature t measured at point p during iteration i

- The temperature $t[p]^{i+1}$ at iteration i+1 is computed as the average of the temperature of point p, its predecessor p-1 and its successor p+1 at iteration i

$$t[p]^{i+1} = (t[p-1]^i + t[p]^i + t[p+1]^i)/3$$

Requirements

- Boundary conditions
  - For the first point t[0] and for the last point t[n-1], consider the average of the temperature of the point and its only neighbor point

$$t[0]^{i+1} = (t[0]^{i} + t[1]^{i}) / 2$$

$$t[n-1]^{i+1} = (t[n-1]^{i} + t[n-1]^{i}) / 2$$

Requirements

- At each round, you have to compute and print the minimum and maximum values of temperature over all points, and the difference between the minimum and the maximum
  - The code for printing is already in the template
- The simulation should stop when the difference between the maximum and the minimum is below a given threshold
  - Variable allowed_diff in the template
- Run the simulation in parallel on multiple processes, minimizing synchronization and communication as much as possible
  - You may assume the number of points to be a multiple of the number of processes involved in the computation